Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/03/2005
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Project Title:	Control of Bacterial Biofilm Formation in Space Exploration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
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	Bacteria in nature and disease often choose a mode of growth known as a biofilm, which consists of numerous aggregating microorganisms (microcolonies) attached to a surface. Biofilm formation is a leading cause of bacterial pneumonia which can be acquired by astronauts during a long-term space mission and by patients with compromised immune systems such as those with cystic fibrosis. A capsular material called mucoid exopolysaccharide, or alginate is important in the formation of biofilms because it provides a key structural component for in vivo biofilms. The objective of this proposal is to study the genetic regulation of alginate in biofilm formation. The model organism is Pseudomonas aeruginosa, a ubiquitous biofilm-forming bacterium capable of displaying two distinctive phenotypes: a nonmucoid or mucoid colony morphology. The mucoid form, which is due to the overproduction of alginate, is a clinical indicator for a poor prognosis. While the two morphologies can be made inter-changeable under the laboratory conditions, little is		

Task Description:	known about the molecular mechanisms governing this process. Using genetics and functional genomics approaches, we will search for the biofilm “switches” on the genomes. The identified genes may become targets for the development of anti-biofilm therapeutics. The specific aims of this project are i) to identify and characterize the genes in the alginate regulation, ii) to elucidate the underlying mechanisms, and iii) determine what/how environmental signals modulate the expression of these regulators. Our long-term goal is to target bacterial own genes for the control of the space biofilm formation. The objective of the proposed research is to identify and characterize the biofilm-controlling genetic elements from the genomes of P. aeruginosa. It is our expectation that the approaches used here will lead us to identify the genes that control the formation of the mucoid biofilms. We will also provide new information regarding how these genes control the phenotypic switch, and what/how environmental signals may stimulate or inhibit the expression of these genes. These results will be significant, because they will provide genetic targets for anti-biofilm therapeutics, and aid in the future development of the gene-based space medicine. In addition, it is expected that the results will advance the emerging field of microbial biofilms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major Earth benefits of this proposal are to provide the basis for the future development of the new genetically defined drug with improved specificity and function and very little side effect.
Task Progress:	1. Elucidation of a signal transduction pathway of alginate regulation. The transposon mutant banks generated for this project provided genetic evidence that AlgU is the master regulator in alginate overproduction. Furthermore, we had insights on how the protein network regulates this phenotype. It is clear that alginate is controlled via a signal transduction system that converges at the transcriptional and post-translational expression of algU. However, the direct evidence on how each component interacts with each other is still lacking. The emphasis for the remaining time of this project will focus on the analysis of the interaction of each player. We will construct the transcriptional fusion using the alginate-specific genes such as algU, rpoN, algR, and algB fused with reporter systems such as gfp or lacZ. To further characterize these mutants, we will test the specific and global effects of the mutation using the real-time PCR, microarray and 2-D gel electrophoresis. 2. Characterization of suppressor mutation that causes the phenotypic reversion from a mucoid to nonmucoid morphology. We will determine the reversion rates of these defined mutants using the Luria-Delbruck fluctuation test. We will continue to search for the original and suppressor mutations in PAO579 and PAO579NM. Once these mutations are found, we will change them from mutant form back to wild type or vice versa to confirm with the phenotype that they are indeed the causative mutations. The identification of these mutations will help to understand how they affect the phenotypic switch from nonmucoid <--> mucoid <--> nonmucoid. 3. Development of an efficient recombination system for gene inactivation in P. aeruginosa. We will develop a P. aeruginosa strain with an increased frequency of homologous recombination using the linear PCR molecules as templates will be useful. One application is the use of this recombinant to investigate the effect of radiation on recombination and DNA repair systems in P. aeruginosa. While current knowledge on the DNA repair system is derived mainly from the studies on E. coli, it would be interesting to see whether there is difference and similarity in terms of the DNA repair and recombination systems when P. aeruginosa is irradiated. It is clear that space irradiation can chemically damage DNA due to its absorption of the energy of the photons. DNA damage can cause mutations, many of which may be deleterious or lethal. Therefore, development of this recombinant strain may allow us to expand our research to a new field of space biology.
Bibliography Type:	Description: (Last Updated: 04/09/2020)
Abstracts for Journals and Proceedings	Skolnick KR, Yu HD. "Interleukin-12 is required for the innate host defense against lung colonization by Pseudomonas aeruginosa." American Society for Microbiology 105th General Meeting, Atlanta, GA, June 2005. American Society for Microbiology 105th General Meeting, June 2005. http://www.asm.org , Jun-2005
Abstracts for Journals and Proceedings	Head NE, Skolnick KR, Yu HD. "Overproduction of alginate in a cystic fibrosis isolate of Pseudomonas aeruginosa results in reduced virulence in a susceptible mouse model." American Society for Microbiology 105th General Meeting, Atlanta, GA, June 2005. American Society for Microbiology 105th General Meeting, June 2005. http://www.asm.org , Jun-2005
Presentation	Yu, H. D. "Interleukin-12 is required for the innate host defense against lung colonization by Pseudomonas aeruginosa" 3rd Mid-Atlantic Microbial Pathogenesis Meeting, Wintergreen Resort, Wintergreen, VA Feb-2005
Significant Media Coverage	Tarbett J. "Cystic Fibrosis Foundation hosting walk to benefit research." Huntington Herald-Dispatch. http://www.herald-dispatch.com/2005/May/16/LFlist1.htm , May-2005